Networkwalks Internship

Wireshark SOC Analyst Project Module

Submitted in partial fulfillment of the practical requirements for the Cybersecurity Internship Programme.

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Prepared by the Intern

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1 Introduction

As part of the Networkwalks Cybersecurity Internship, I was tasked with completing six live SOC (Security Operations Center) labs using Wireshark. The objective was to step into the role of a Tier 1 SOC Analyst, capture live network traffic, and investigate various security incidents.

The core focus of this module was to bridge the gap between theoretical networking concepts and practical incident response. The labs covered a spectrum of threats, including active reconnaissance (port scanning), credential interception, brute-force attacks, data exfiltration via cleartext protocols, and VoIP call eavesdropping.

2 Laboratory Methodology

For each lab, the following general workflow was applied:

1. **Environment Setup:** Logged into the Windows 11 Virtual Machine and launched Wireshark on the WiFi interface.
2. **Traffic Capture:** Captured the traffic until the lab scenario's stimulus had concluded.
3. **Filtering and Analysis:** Applied specific display filters to isolate malicious or anomalous traffic from normal background noise (e.g., ARP, DNS, and standard web browsing).
4. **Evidence Extraction:** Used Wireshark's "Follow TCP Stream" and packet detail panes to reconstruct application-layer payloads and extract answers.

3 Lab 1: Identify the Attacker's IP

Scenario: Unusual traffic was reported to a web server. Identify the attack type and the attacker's IP.

Technical Workflow

- Applied the filter `tcp.flags.syn == 1` to display only TCP connection requests.
- Observed a single IP address (`192.168.10.66`) sending a rapid, sequential block of SYN packets to the target server (`192.168.10.20`).
- Analyzed the response packets: open ports replied with `[SYN, ACK]` (Ports 22, 80, 443), while closed ports replied with `[RST, ACK]`.

Findings

- **Attacker IP:** `192.168.10.66`
- **Target IP:** `192.168.10.20`
- **Ports Probed:** 40 (3 open, 37 closed).
- **Attack Type:** Port Scan (Active Reconnaissance).

4 Lab 2: Follow TCP Stream (Part 1)

Scenario: A staff member used an unencrypted office chat tool. Read the conversation and identify what sensitive data was shared.

Technical Workflow

- Used `Statistics > Conversations > TCP` to identify non-standard ports. Found an active chat on **Port 5000**.
- Filtered for `tcp.port == 5000` and used **Follow ¿ TCP Stream**.
- Read the cleartext conversation between the client (Ali) and the IT server.

Findings

- **Client:** Ali (Accounts department, IP: `192.168.10.45`)
- **Temporary Password Leaked:** `Tiger#4821`
- **Guest Wi-Fi Password Leaked:** `BlueHarbor77`

- **Key Takeaway:** Cleartext chat applications are highly vulnerable to passive network sniffing.

5 Lab 3: Follow TCP Stream (Part 2)

Scenario: An admin believes a secret was left inside a web page on the company intranet.

Technical Workflow

- Filtered for `http` to see all page requests.
- Right-clicked and followed TCP streams for the various pages requested.
- Located a hidden note embedded in the HTML code of `/staff.html`.

Findings

- **Hidden Data Location:** An HTML Comment (`<!-- ... -->`) in Stream 7.
- **Code Word:** `Falcon-9921`
- **Backup Server IP:** `192.168.10.77`
- **Key Takeaway:** Browsers hide HTML comments from users, but network sniffers can see them in transit.

6 Lab 4: HTTP Plain Text Credentials

Scenario: A staff member signed into an old web portal. Intercept the login and extract the password.

Technical Workflow

- Applied the filter `http.request.method == "POST"` to find form submissions.
- Identified two login attempts. Clicked the second packet.
- Navigated to the packet details pane: `HTML Form URL Encoded`.

Findings

- **Protocol:** HTTP (Unencrypted).
- **Username:** `farhan.k`

- **Successful Password:** `Mango#2026` (First attempt with `Mango2025` failed).
- **Key Takeaway:** HTTP transmits data in plain text. HTTPS is mandatory for authentication.

7 Lab 5: Catch the Brute Force Attacker

Scenario: An FTP server owner reports many failed logins. Identify the attacker and the successful password.

Technical Workflow

- Used `Statistics > Protocol Hierarchy` to confirm FTP usage.
- Filtered for `ftp.request.command == "USER"` and `"PASS"`.
- Analyzed failure codes (`530`) and the success code (`230`). Used Follow TCP Stream on the success packet to see the password.

Findings

- **Attacker IP:** `192.168.10.66`
- **Target IP:** `192.168.10.20`
- **Failed Attempts:** 24
- **Successful Password:** `Sunshine99`
- **Attack Type:** Brute Force (Password Guessing).

8 Lab 6: VoIP Call Replay

Scenario: A phone call was intercepted. Replay the call and identify the sensitive information shared.

Technical Workflow

- Navigated to `Telephony > VoIP Calls`.
- Selected the identified SIP call and clicked **Play Streams**.
- Listened to the audio playback to extract the information verbally shared.

Findings

- **Setup Protocol:** SIP (Port 5060)
- **Voice Protocol:** RTP (Codec: PCMU)
- **Caller:** Ali Raza (Extension 101) / **Receiver:** Sara (Extension 202)
- **Credit Card Number:** `4213 5588 7710 2394` (Expiry: 09/28)
- **Key Takeaway:** Voice packets (RTP) are unencrypted by default. SRTP and TLS must be used to prevent eavesdropping.

9 Conclusion and Reflection

Completing these six labs provided invaluable hands-on experience in network forensics and incident response. Transitioning from theoretical knowledge of the OSI model to actively filtering live PCAP data was challenging but highly rewarding.

Key Technical Takeaways:

- **Mastery of Wireshark Filters:** Understanding how to quickly isolate specific traffic (e.g., `tcp.flags.syn == 1`, `http.request.method == "POST"`) is critical for speed during an incident.
- **The Danger of Cleartext:** The HTTP, FTP, and Chat labs clearly demonstrated how devastating a lack of encryption is. A simple network sniffer can capture passwords, financial data, and private conversations.
- **Reconstruction:** The "Follow TCP Stream" feature is a powerful tool for reconstructing the application layer narrative from raw packets.
- **Encryption is Non-Negotiable:** Whether it is web traffic (HTTPS), file transfer (SFTP), or voice calls (SRTP), encryption is the only way to protect data in transit.

This module has significantly improved my confidence in analyzing network traffic and hunting for threats. I look forward to applying these skills in a real-world SOC environment.